

# Systems of Ordinary Differential Equations

## Problem 1

“Flattening the curve” became a global rallying-cry during the early stages of the COVID-19 pandemic. In this problem we explore this phrase in the context of the SIR model. Recall that the SIR model for disease transmission involves three populations.

- $S(t)$ : the number of susceptible people  $t$  days after the outbreak.
- $I(t)$ : the number of infected people  $t$  days after the outbreak.
- $R(t)$ : the number of removed people  $t$  days after the outbreak (recovered or deceased).

The evolution of these populations is described by the following system of ordinary differential equations.

$$\begin{cases} \frac{dS}{dt} = -aS(t)I(t), \\ \frac{dI}{dt} = aS(t)I(t) - bI(t), \\ \frac{dR}{dt} = -\frac{dS}{dt} - \frac{dI}{dt} = bI(t). \end{cases}$$

- (a) When people speak of “flattening the curve” which of these three curves do they most directly mean to flatten?
- (b) Consider each of the following measures taken to slow the spread of COVID-19, and indicate whether they increase or decrease the constants  $a$  and  $b$ .
  - (a) Developing medicines to speed up recovery.
  - (b) Enacting social distancing.
  - (c) Encouraging all people to wear a face-mask in public.

*Solution.*

- (a) “Flattening the curve” refers to the curve of infected people over time,  $I(t)$ .
- (b) Recall that  $a$  is the daily rate of infection, and it is the reciprocal of the average number of days  $\gamma$  that an individual remains infected,  $a = \gamma^{-1}$ . On the other hand,  $b$  is the daily rate of removal and it is the product of the fraction  $\alpha$  of the daily possible encounters between susceptible and infected that actually occur and the fraction  $\beta$  of these that result in contagion,  $b = \alpha\beta$ .
  - (a) Developing medicines to speed up recovery would decrease  $\gamma$  and therefore increase  $b$ . It would not affect  $a$ .
  - (b) Enacting social distancing would decrease  $\alpha$  and therefore  $a$ . It would not affect  $b$ .
  - (c) Face-masks would decrease  $\beta$  and therefore  $a$ . They would not affect  $b$ . ■

**Problem 2**

The system of differential equations

$$\begin{cases} x'(t) = -3x + 6xy, \\ y'(t) = y - 4xy, \end{cases}$$

is a model for a population consisting of foxes (predators) and hares (prey).

- Identify which variable,  $x$  or  $y$ , represents the fox population and which variable represents the hare population.
- Find the equilibrium points (constant solutions) for the system of differential equations and explain their significance.
- Find a single differential equation whose solutions give the fox population as a function of the hare population and solve it.

*Solution.*

- The quantity  $xy$  corresponds to the number of interaction between the foxes and the hares. In the equation for  $x'$ , this term appears with a positive sign. This means that population  $x$  benefits from increased interactions; it must be the foxes. On the other hand, in the equation for  $y'$ , the interaction term appears with a negative sign. This means that population  $y$  suffers from increased encounters; it must be the hares.
- The equilibrium points  $(x, y)$  are constant solutions to the system of ordinary differential equations, and must therefore satisfy

$$\begin{cases} -3x + 6xy = 0, \\ y - 4xy = 0. \end{cases}$$

The only solutions to this system of equations are  $(x, y) = (0, 0)$  and  $(x, y) = (1/4, 1/2)$ . The significance of these values is that, if the populations ever attain either of these states, they will remain there indefinitely.

- Using the chain rule reveals that

$$\frac{dx}{dy} = \frac{dx/dt}{dy/dt} = \frac{x'(t)}{y'(t)} = \frac{-3x + 6xy}{y - 4xy} = \frac{x(6y - 3)}{y(1 - 4x)}.$$

This is a separable differential equation. Cross multiplying and integrating yields

$$\int \frac{1 - 4x}{x} dx = \int \frac{6y - 3}{y} dy.$$

This means that  $x$  and  $y$  are related through the implicit equation

$$\ln(x) - 4x = 6y - 3\ln(y) + C. \quad \blacksquare$$